

CS 5/7320

Artificial Intelligence

Introduction to AI

AIMA Chapters 1 + 27

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based on slides by Svetlana
Lazepnik with figures and cover
art from the AIMA textbook.



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Stuart
Russell
Peter

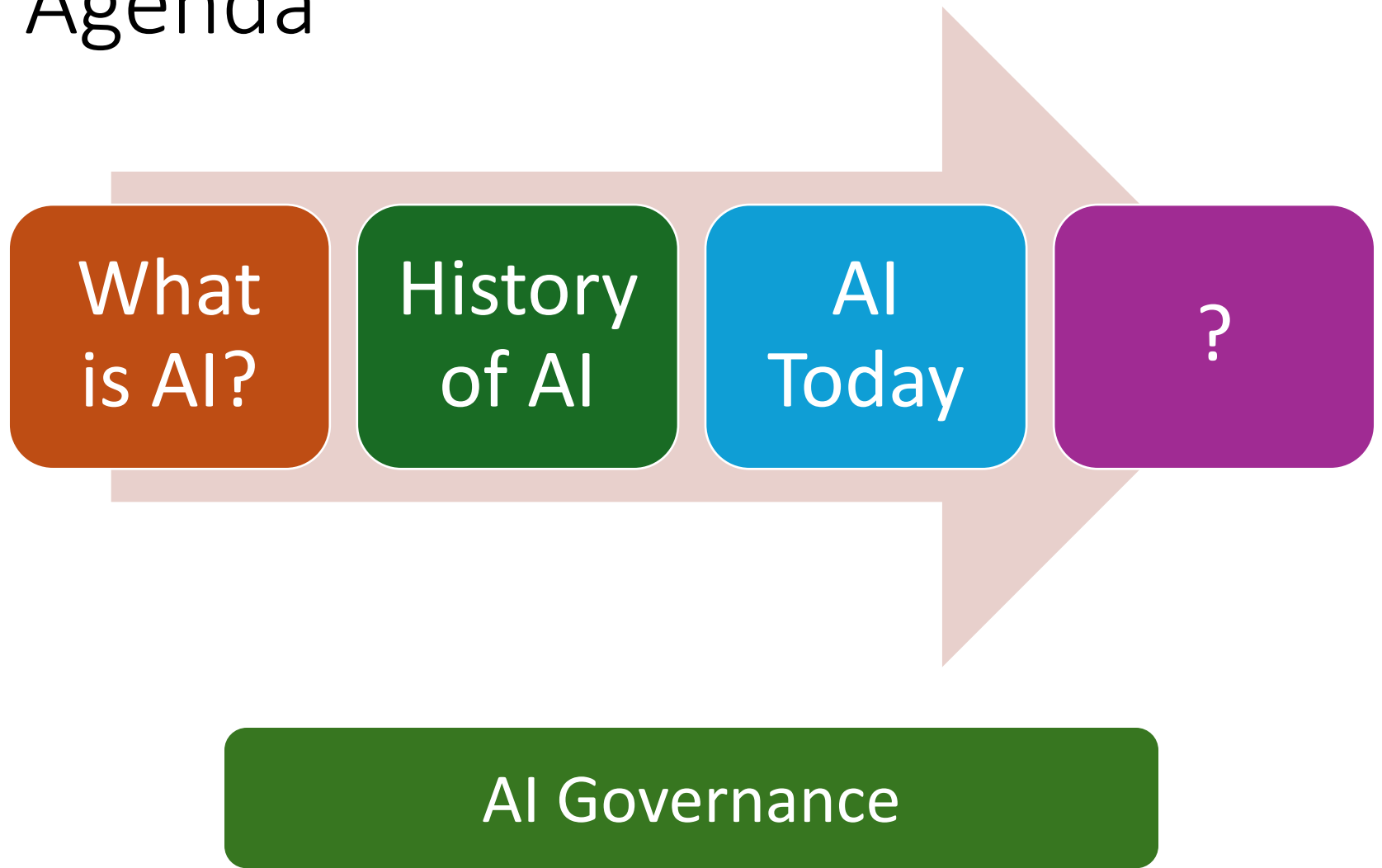
Artificial Intelligence
A Modern Approach



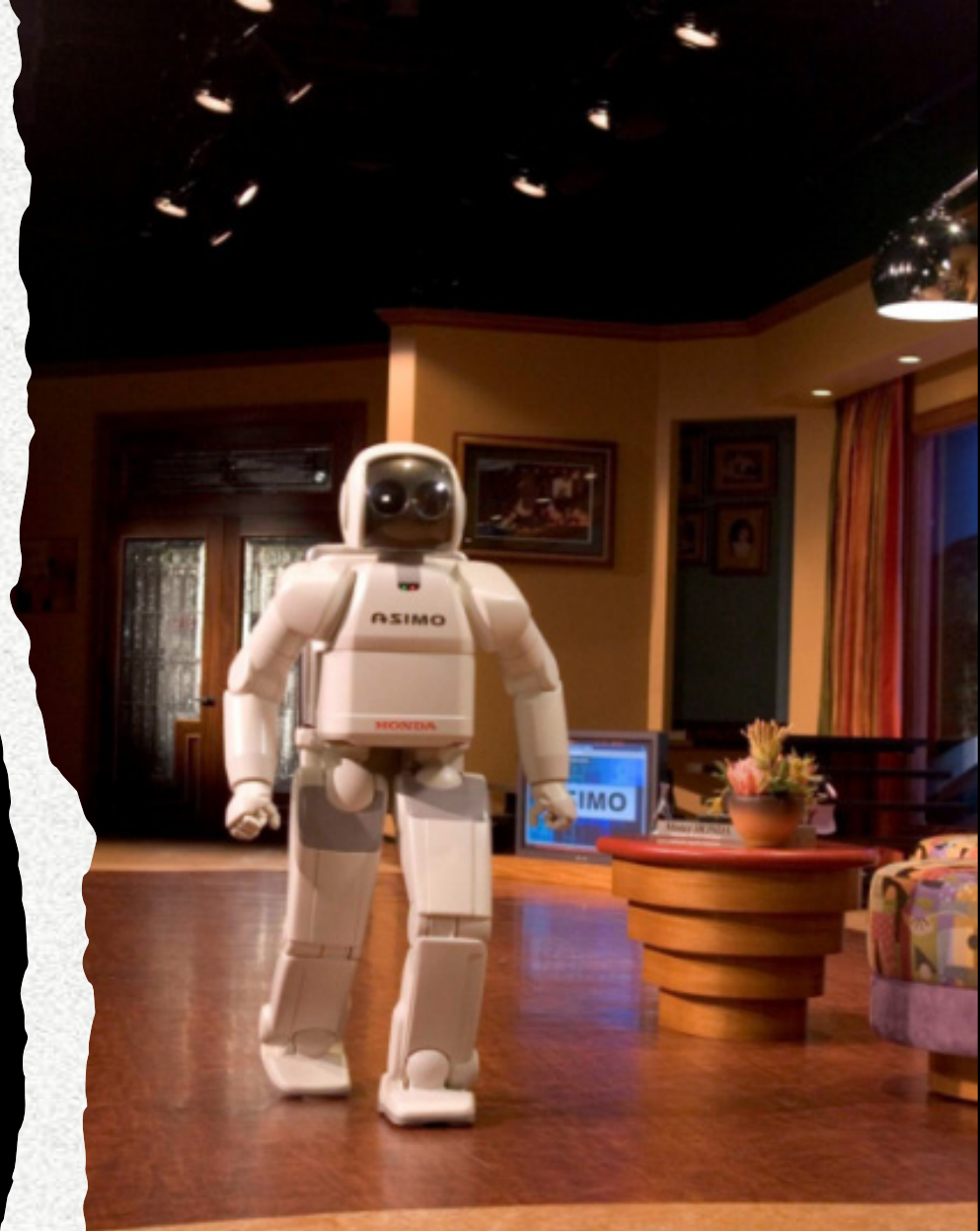
Online Material

Fourth Edition

Agenda



What is Artificial Intelligence?



ASIMO (Advanced Step in Innovative Mobility)
is a humanoid robot created by Honda in 2000

What is Intelligence?

Definition from Merriam-Webster

intelligence **noun**

in·tel·li·gence (in-ˈte-lə-jən(t)s)

[Synonyms of intelligence >](#)

1 a (1) : the ability to learn or understand or to deal with new or trying situations :

REASON

also : the skilled use of reason

(2) : the ability to apply knowledge to manipulate one's environment or to think abstractly as measured by objective criteria (such as tests)

b : mental acuteness : **SHREWDNESS**

c **Christian Science** : the basic eternal quality of divine Mind

2 a : **INFORMATION, NEWS**

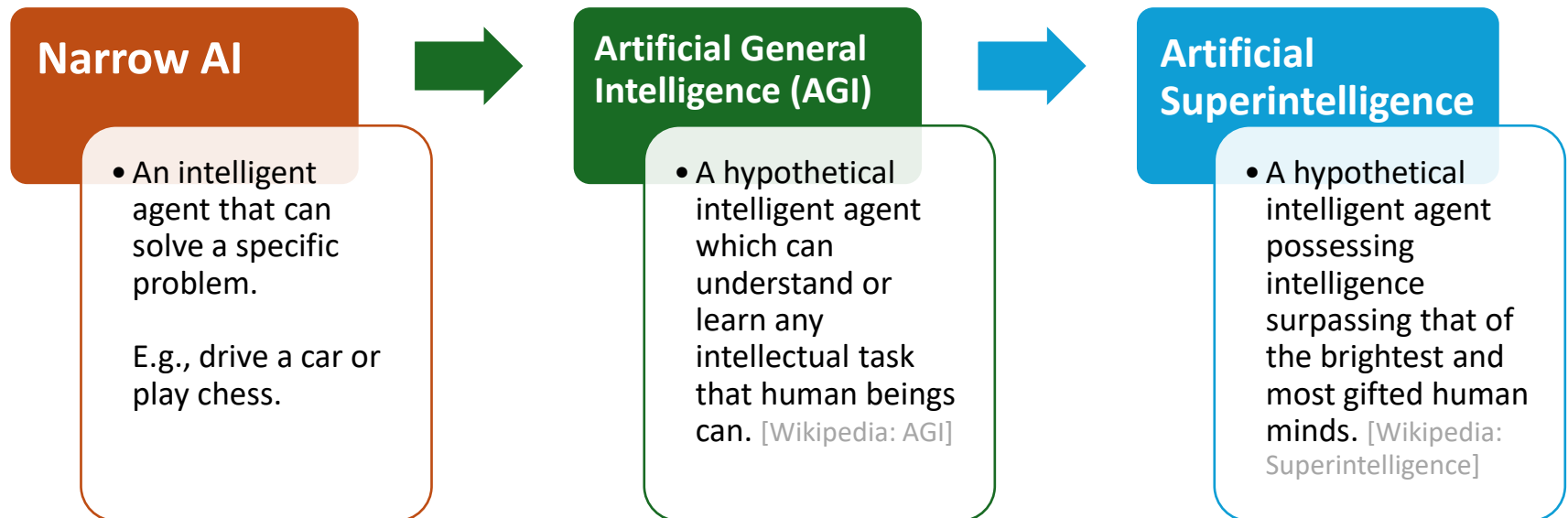
b : information concerning an enemy or possible enemy or an area

also : an agency engaged in obtaining such information

The Goal of AI

“Have machines solve problems that are challenging for humans.”

We call such a machine an **intelligent agent**.



How can we achieve this? Create an agent that can:

Think like a human?

Act like a human?

Think rationally?

Act rationally?

Think like a
human?

Act like a
human?

Think
rationally?

Act
rationally?

The brain as an
information processing
machine.

- Requires scientific theories of **how the brain works**.

Note: The brain does not work like artificial neural networks from machine learning!

How to understand
cognition as a
computational process?

- Introspection: try to think about how we think.
- **Predict the behavior of human** subjects.
- Image the brain, examine neurological data

AI consciousness

- What does it mean that a machine is conscient/sentient?
- How can we tell?
- What do we do then?



Cognitive Sciences

**Moral
Philosophy**

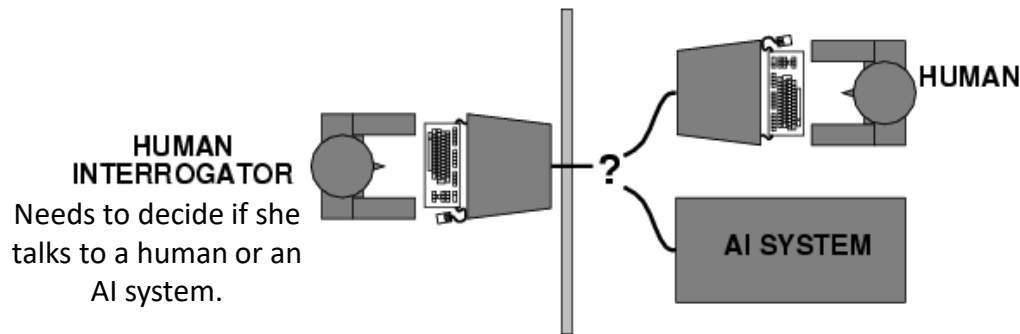
Think like a
human?

Act like a
human?

Think
rationally?

Act
rationally?

- Alan Turing rejects the question “Can machines think?”
- The Turing Test tries to define what acting like a human means.



Alan Turing (1950)
“Computing
machinery and
intelligence”

- What capabilities would a computer need to have to pass the Turing Test?
These are still the core AI areas.
 - Natural language processing
 - Knowledge representation
 - Automated reasoning
 - Machine learning
- Turing predicted that by the year 2000, machines would be able to fool 30% of human judges for five minutes.
ChatGPT (2023) is probably doing a least that!

Turing Test: Criticism

What are some potential problems with the Turing Test?

- Some human behavior is not intelligent.
- Some intelligent behavior may not be human.
- Human observers may be easy to fool.
 - A lot depends on expectations.
 - Anthropomorphic fallacy: humans tend to humanize things.
- Imitate intelligence without intelligence. E.g., the early chatbots ELIZA (1964) simulates a conversation using pattern matching.



Chinese Room Argument



Thought experiment by John Searle (1980): Imitate intelligence using rules.

Is passing the Turing test a good scientific goal?

- Engineering perspective: Imitating a human is not a good way to solve practical problems.
- We can create useful intelligent agents without trying to imitate humans.

Think like a
human?

Act like a
human?

Think
rationally?

Act
rationally?

- **Thinking Rationality:** Draw sensible conclusions from facts, logic and data.
- **Logic:** A chain of argument that always yield correct conclusions.
E.g., “Socrates is a man; all men are mortal; therefore, Socrates is mortal.”
- **Logic-based approach to AI:** Describe a problem in formal logic notation and apply general deduction procedures to solve it.
Issues:
 - Describing real-world problems and knowledge using logic notation is hard.
 - Computational complexity of finding the solution.
 - Much intelligent or “rational” behavior in an uncertain world cannot be defined by simple logic rules.

Example: What about the logical implication

study hard $\Rightarrow$ A in my AI course

Should it rather be

study hard AND be lucky AND ... $\Rightarrow$ A in my AI course

Think like a
human?

Act like a
human?

Think
rationally?

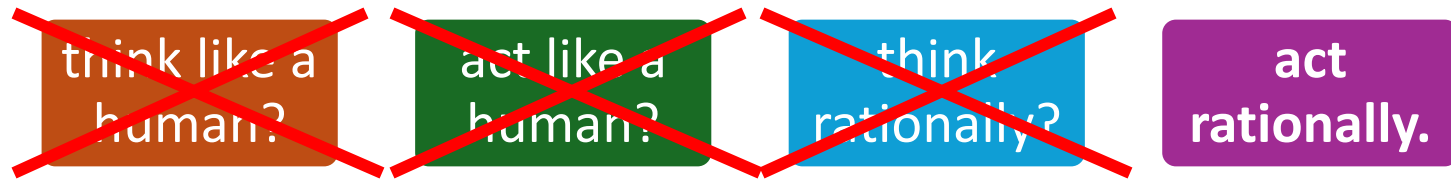
Act
rationally?

Acting rational means to try to
achieve the “best” outcome.

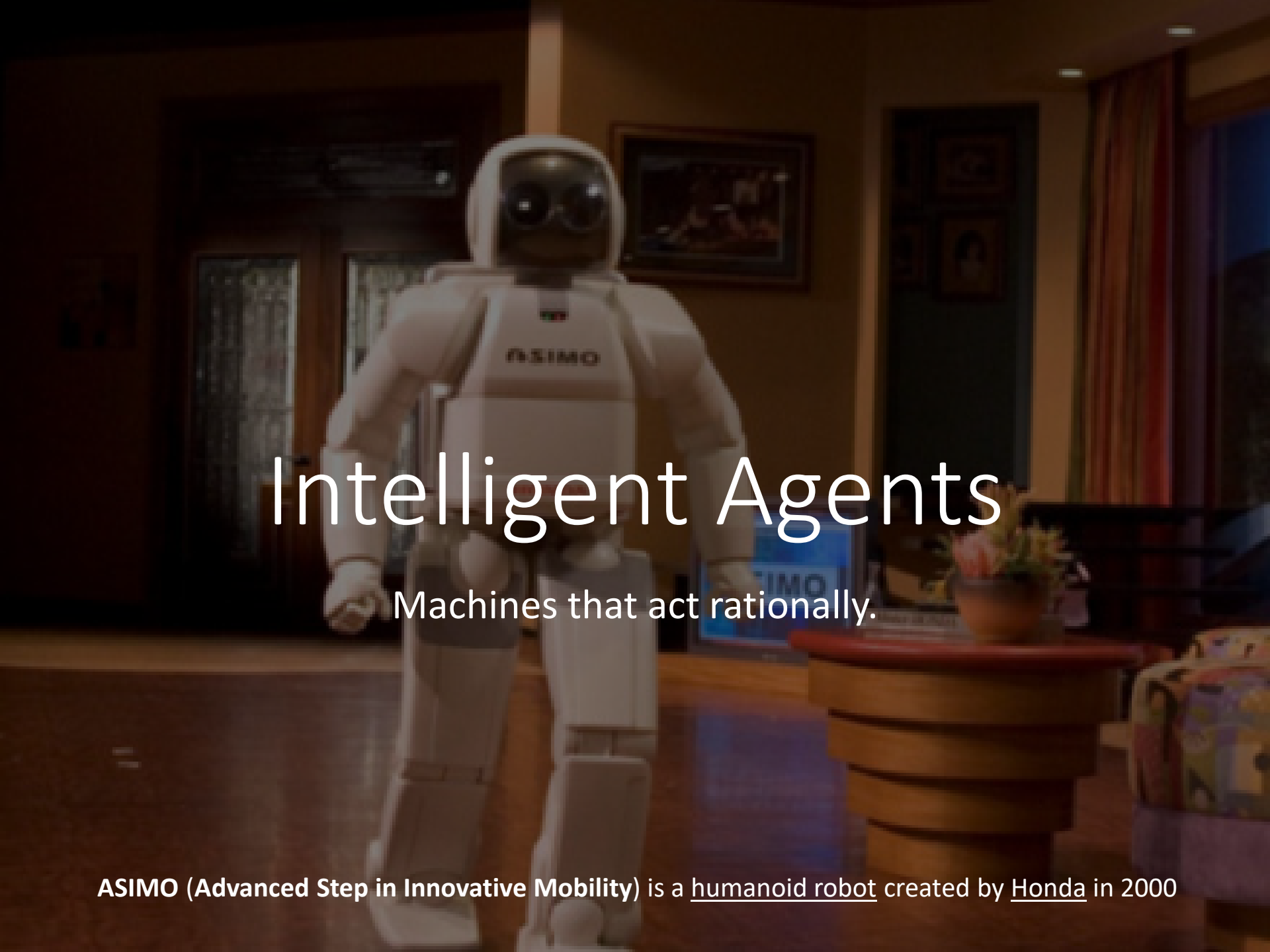
- Best means that we need to do **optimization**.
- The desirability of outcomes can be measured by the economic concept of **utility**. If there is uncertainty about achieving outcomes, then we need to maximizing the **expected utility**.
- Optimization has several advantages:
 - **Generality**: optimization is not limited to logical rules.
 - **Practicality**: can be adapted to many real-world problems.
 - **Well established**: existing solvers and methods for simulation and experimentation.
 - Avoids philosophy and psychology in favor of a **clearly defined objective**.
- **Bounded rationality**: In practice, expected utility optimization is subject to the agent’s knowledge and computational constraints. The agent needs to do the best with its knowledge and resources.

The Type of AI is Covered in this Course

This course focuses on the computational techniques needed to creating a **narrow AI agent** that can



That is, create a machines that acts in a way to solve a specific hard problem that traditionally would have been thought to require human intelligence.

A white humanoid robot, ASIMO, stands in a dimly lit living room. The robot has a friendly face with large, dark eyes and a small smile. It is wearing a white suit with "ASIMO" printed on the chest. The background shows a cozy interior with a fireplace, framed pictures, and a small table with a plant.

Intelligent Agents

Machines that act rationally.

ASIMO (Advanced Step in Innovative Mobility) is a humanoid robot created by Honda in 2000

Components of an Intelligent Agent

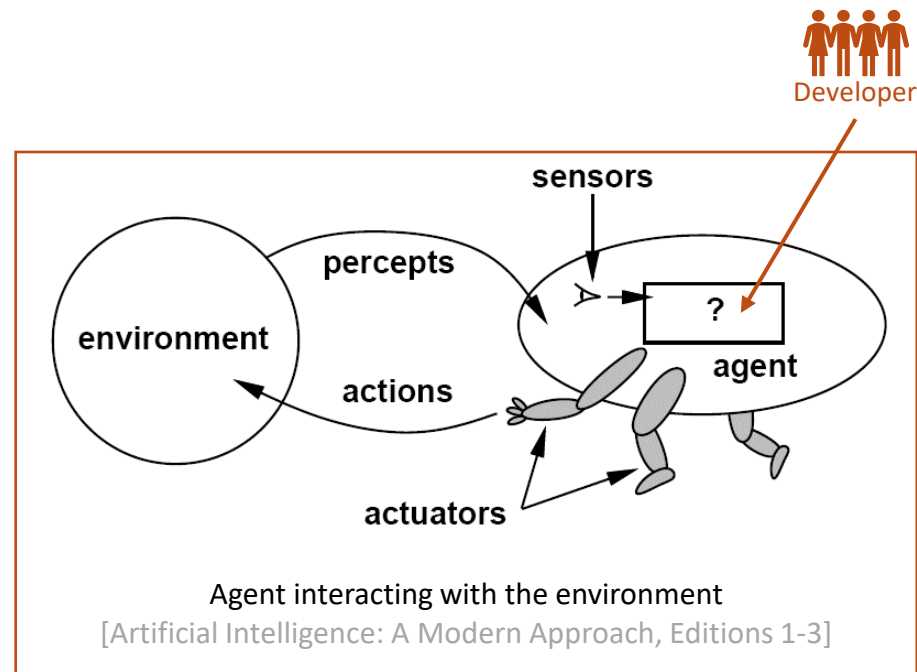
Intelligent agents **act rationally** in their environment.

They need to

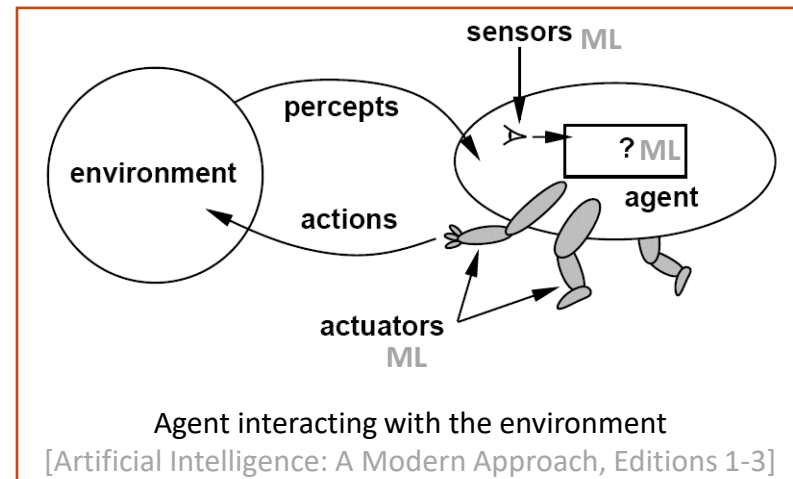
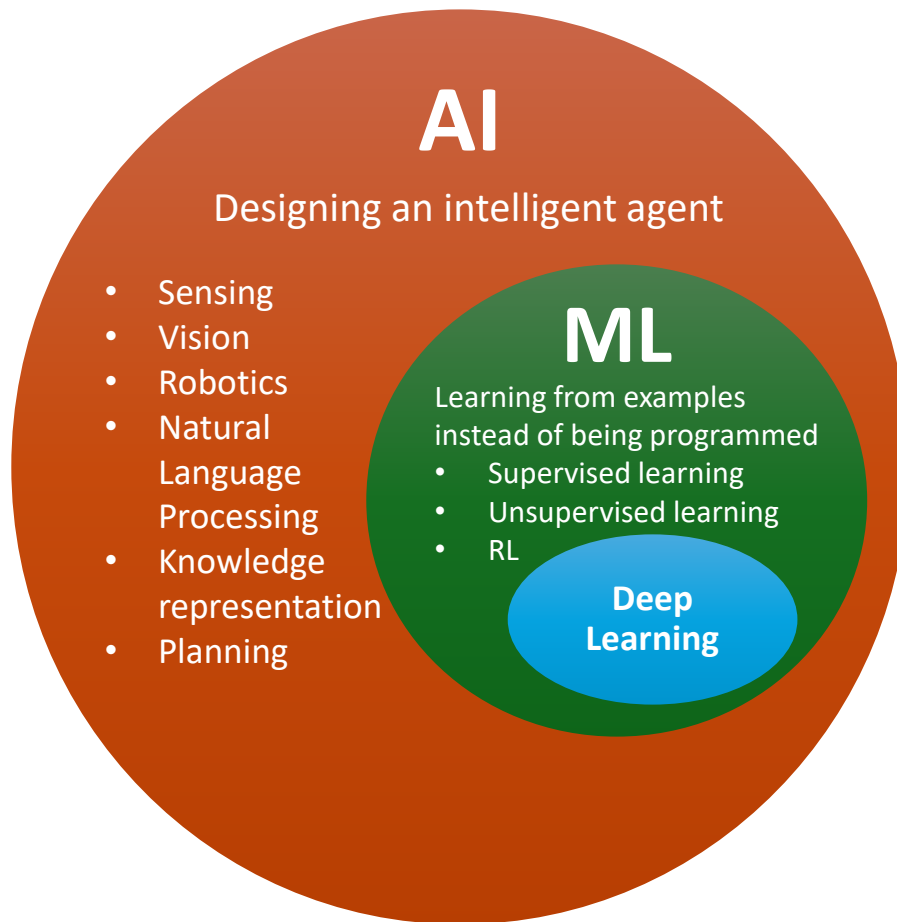
- **Communicate** with the environment using **percepts** and **actions**.
- **Represent knowledge, reason** and **plan** to achieve a desired **outcome**.

Optional

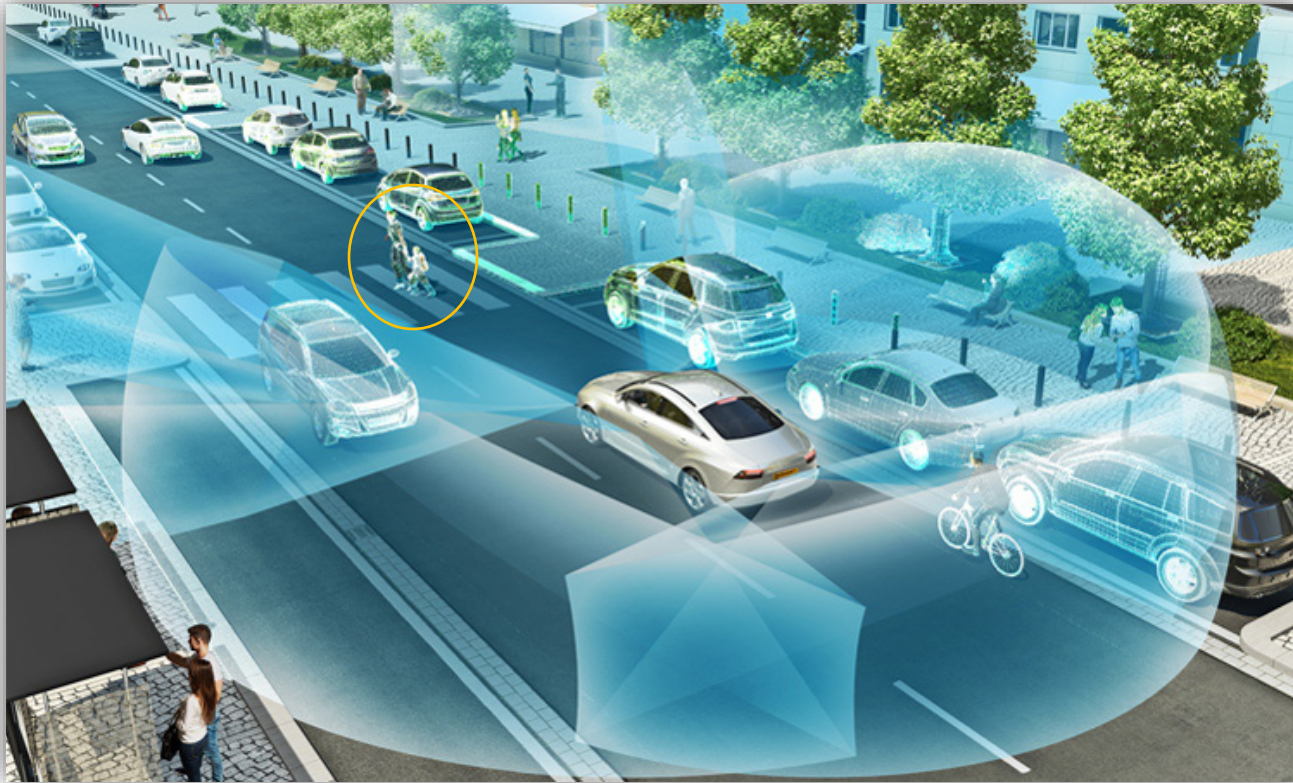
- **Learn from experience** to improve performance.



Artificial Intelligence vs. Machine Learning



Example: Self-Driving Car

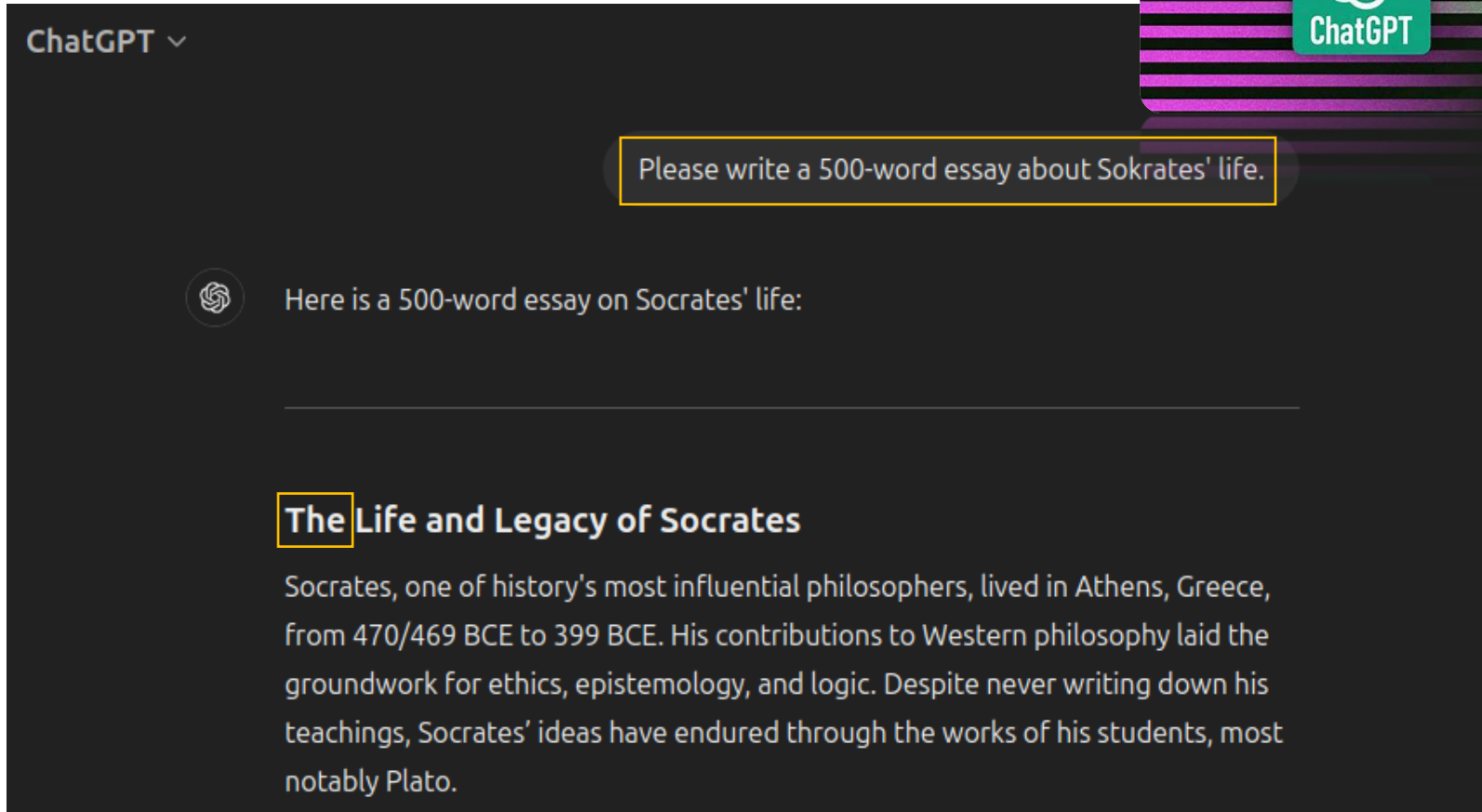


Percepts: Other objects including people crossing the street.

Action: Slow down or stop the car.

Objective: Reach the destination safely

Example: Homework and LLMs



Percept: Your prompt

Action: Next most likely word... More words are created word-by-word.

Objective: You may like a useful answer, but what is ChatGPT's objective?

Methods of AI



Symbolic AI (classical AI)

Store information as explicit facts, objects, and logical rules. Often engineered by humans.

Reasoning

- Search, planning
- Bayesian reasoning
- Classical machine learning (e.g., decision trees)

+



Subsymbolic AI

Operate below the level of individual symbols. Store information as small mathematical equations (weights). Train on massive data sets.

Perception

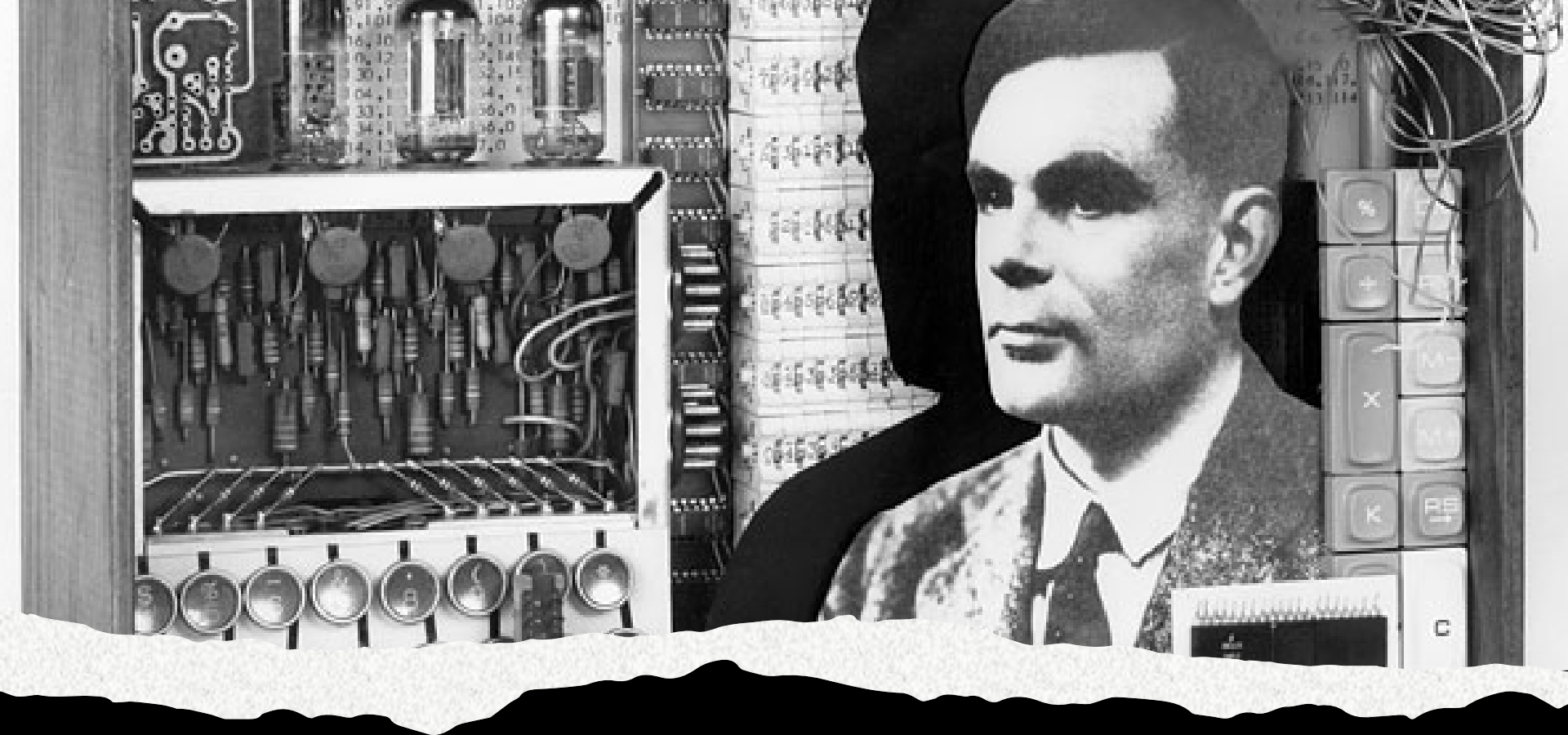
- Artificial neural nets
- Deep learning

||

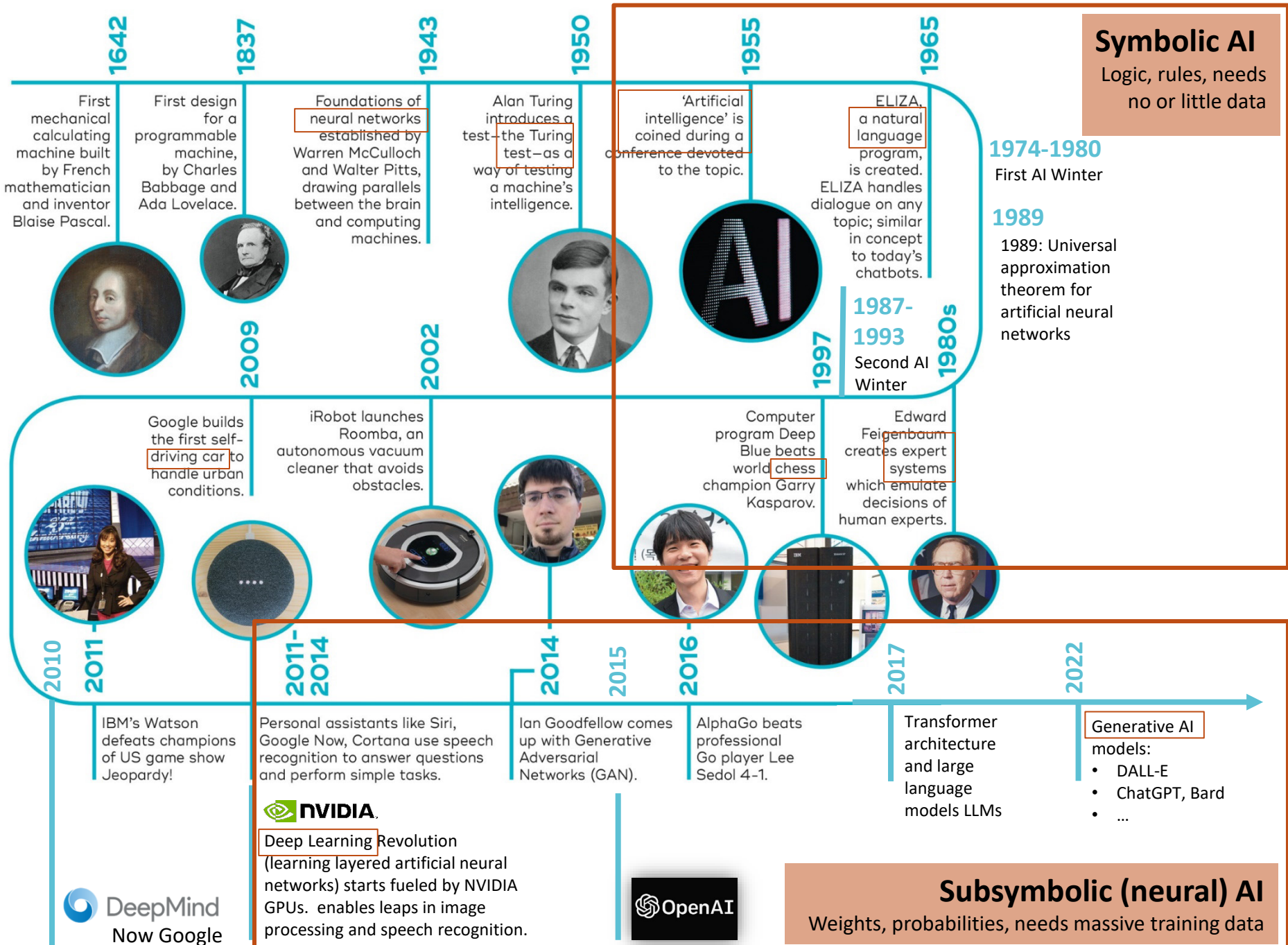


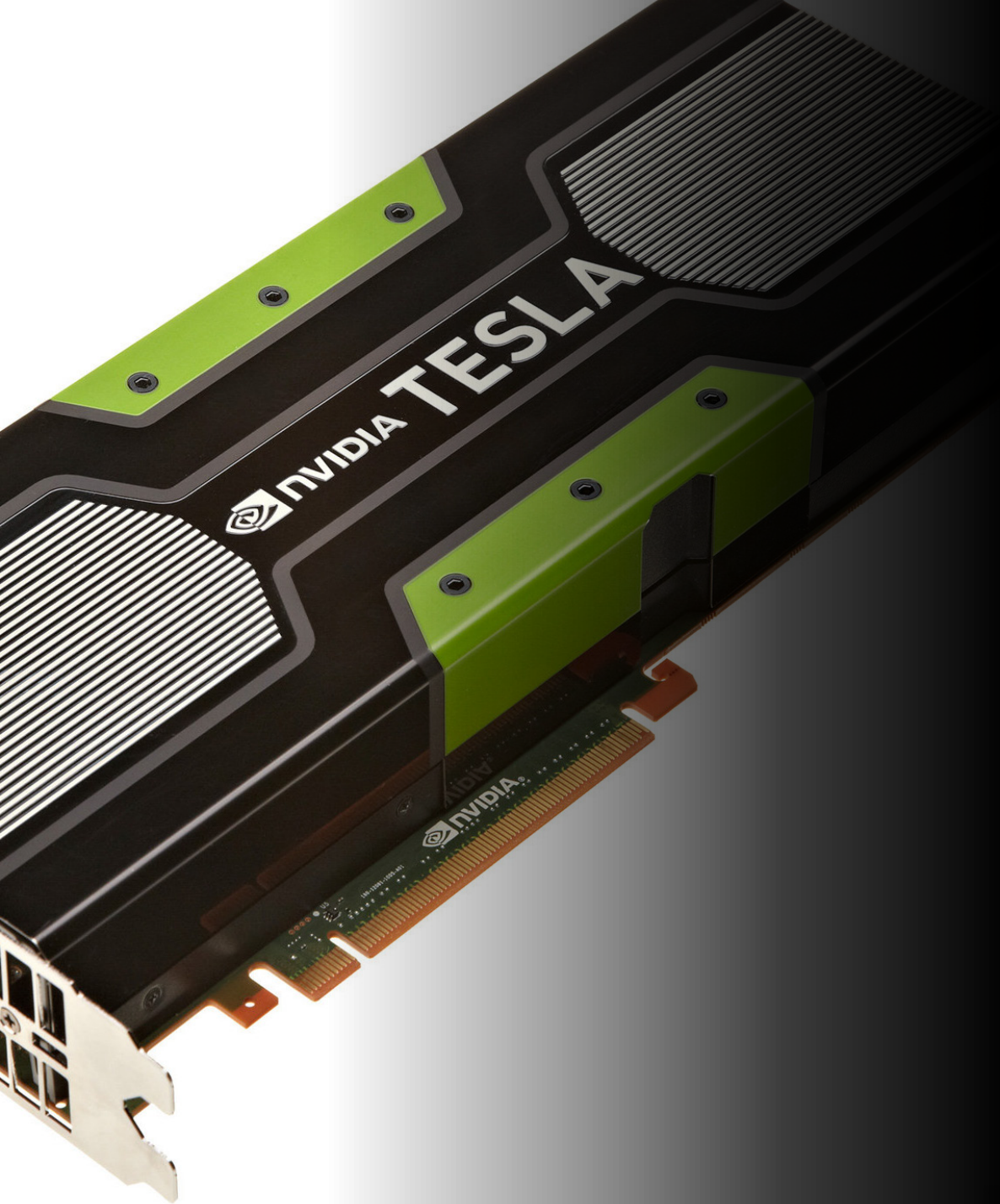
Neurosymbolic (modern) AI

combines neural networks with planning and search.



The History of AI





What accounts for recent successes in AI?

- Dominance of machine learning.
- Faster computers and specialized hardware (GPUs).
- Lots of data (the Internet, text, sensors) and storage (cloud)
- New optimization methods (deep learning).

“Moravec’s Paradox”

Hans Moravec (1988): *“It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and **difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility.**”*

A teenager can learn how to drive in a few hours with very little input, but we still have no truly self-driving car.





The AI Effect: AI gets no respect?

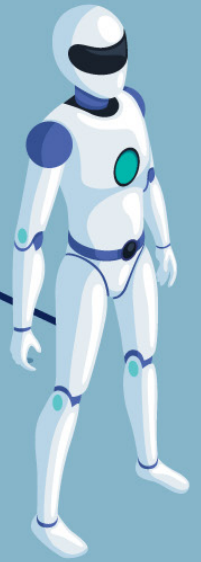
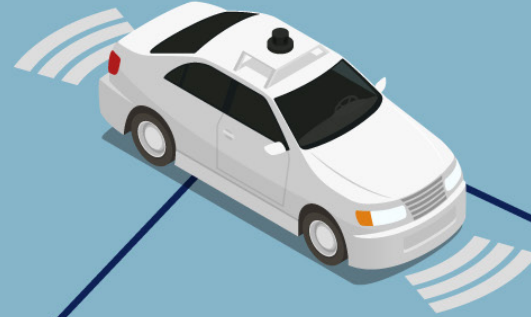
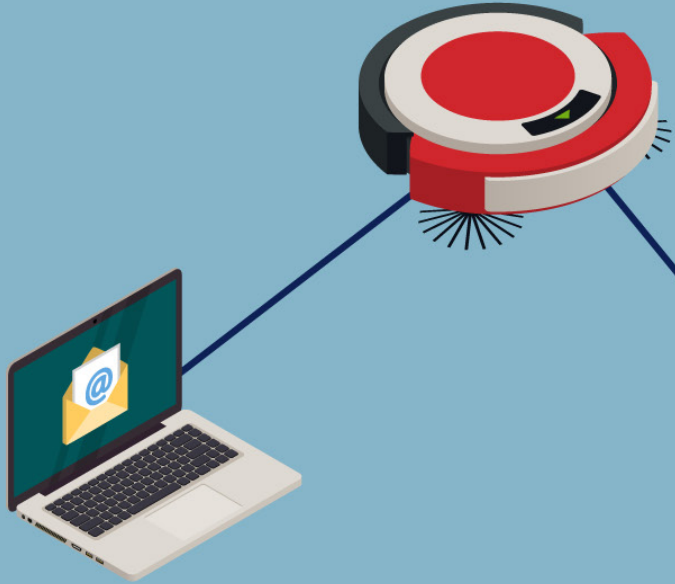
As soon as a machine gets good at performing some task, the task is no longer considered to require much intelligence.

- **Calculating** ability used to be prized – not anymore.
- **Chess** was thought to require high intelligence – now computers play at a super-human level.
- **Learning** once thought uniquely human - now machine learning is a well-developed discipline.
- **Art?** “Even a monkey can do this!”

What about Large Language Models
and Generative AI?

ROOMBA

SELF-DRIVING CAR



AI ROBOT

MAIL SPAM FILTER

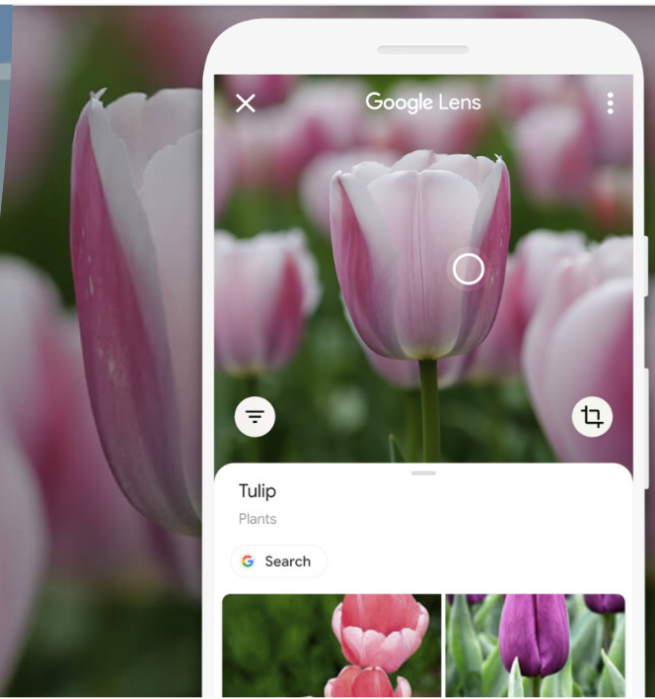
SMART

AI Today

Vision and Image Processing

- **OCR:** read license plates, handwriting recognition (e.g., mail sorting).
- **Face detection:** now standard for smart phone cameras.
- **Vehicle safety systems**
- **Visual search**
- **Image generation**

All these technologies operate now at superhuman performance.



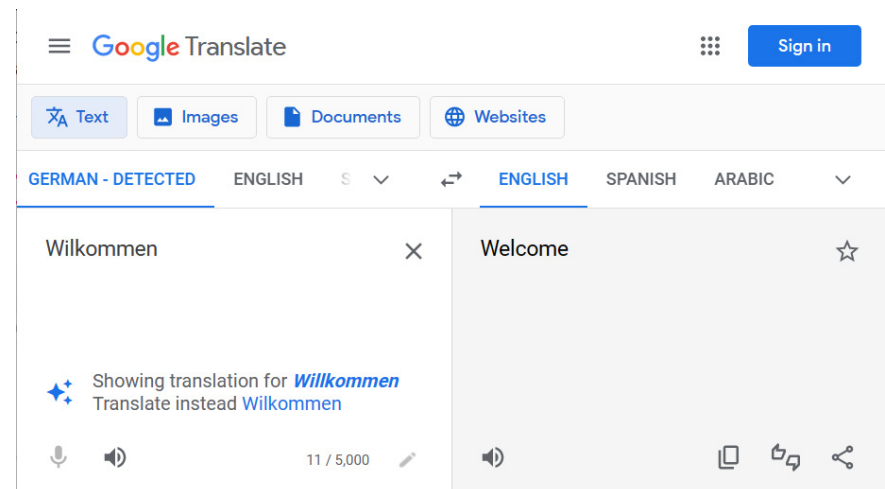
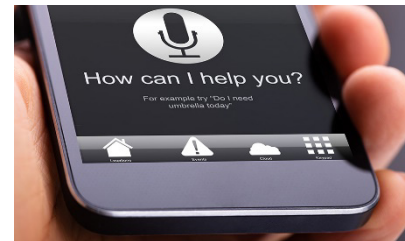
Natural Language Processing

- Text-to-speech
- Speech-to-text to detect voice commands
- Machine translation
- Text generation (Q/A systems) using Large Language Models

These technologies operate now with close to or even superhuman performance.

Humans use language to reason. Does that mean AI that can create high-quality text can reason?

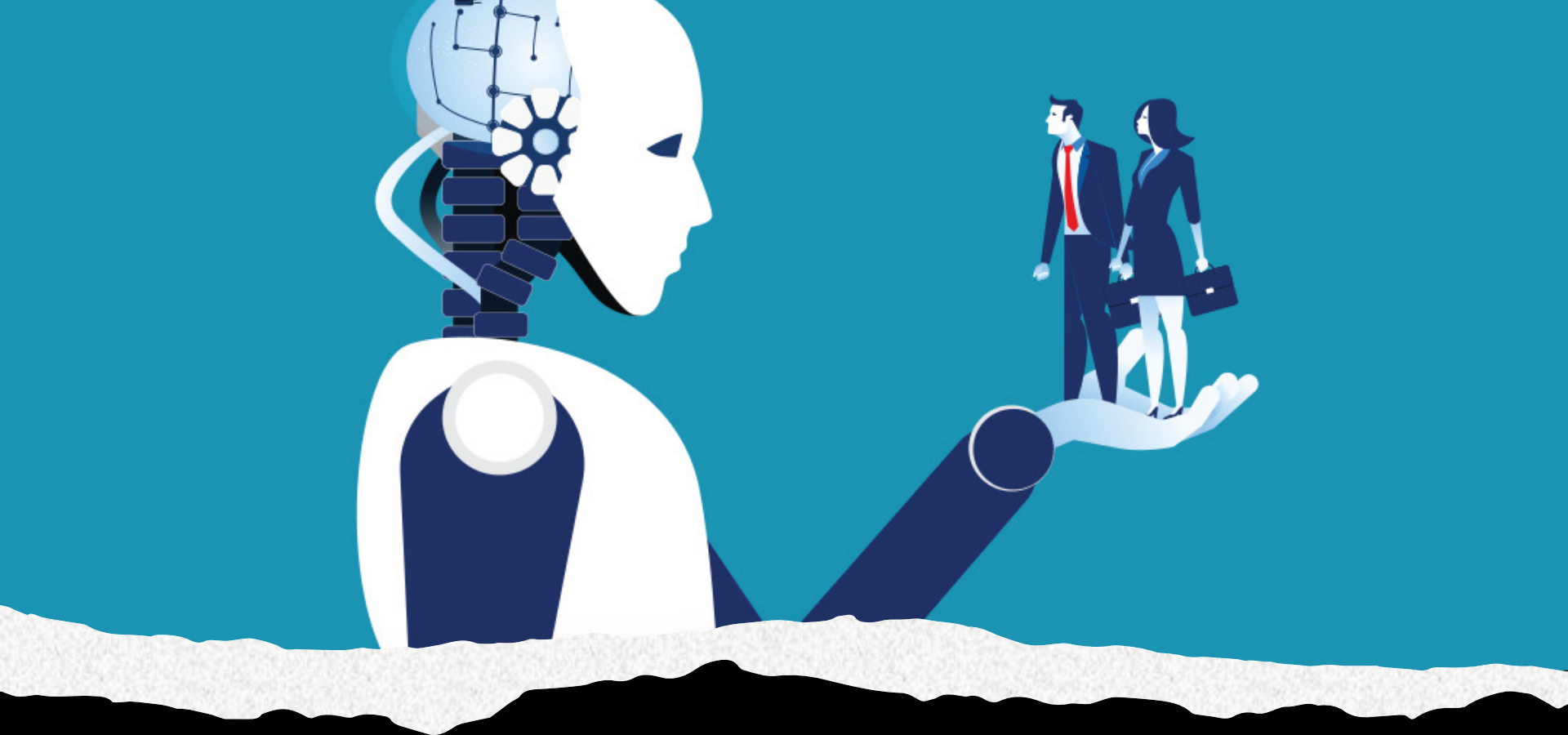
Language understanding is still elusive!



Robotics

- Mars rovers
- Robot soccer
 - [RoboCup](#)
- Autonomous vehicles
 - [DARPA Grand Challenge](#)
 - Self-driving cars
- Drones
- Personal robotics
 - Humanoid robots
 - [Robotic pets](#)
 - Personal assistants?





AI Governance: Ethics & Safety

A new Frontier for
Fairness and Freedom
AIMA Chapter 27 +
online sources

AI Governance

Policies, oversight processes, and operational standards guiding the **ethical, legal, and responsible use of artificial intelligence**.

Key Pillars

Accountability

Clearly defined roles assigning responsibility for AI-driven decisions, system failures, and human oversight checkpoints.

Transparency

The ability to understand how an AI model makes decisions, fostering trust among end-users and regulators.

Fairness

Protocols designed to ensure AI systems do not discriminate against protected groups or yield inequitable outcomes.

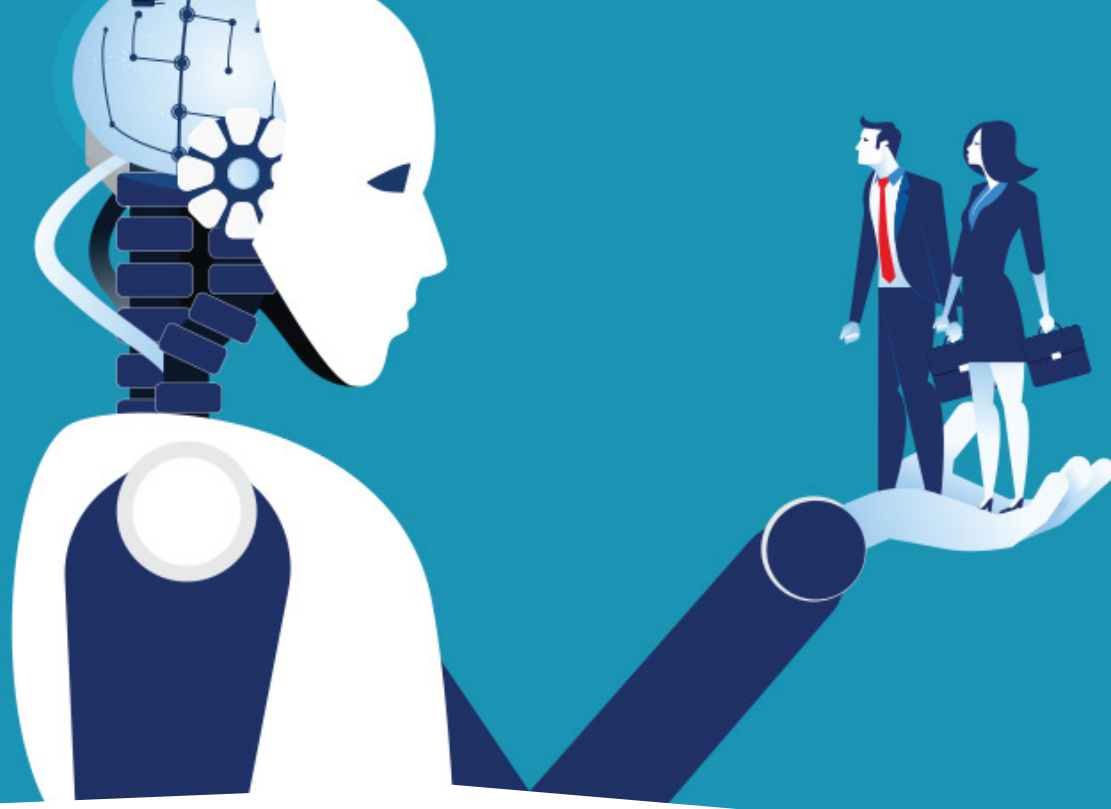
Data Privacy & Security

Strict safeguards dictating how personal and proprietary data is collected, anonymized, stored, and utilized during model training.

Components of AI Governance

Stakeholder trust is created by demonstrating to employees, customers, and investors that AI tools are safe, secure, and beneficial.

- **Compliance:** Follow legislation/guidelines across various jurisdictions to avoiding massive regulatory penalties.
- **Risk Mitigation:** Plan for how to deal with risks involving transparency, fairness, privacy and safety.



AI Governance: Compliance



European Union

Has regulations since 2016 included in the General Data Protection Regulation (GDPR)

[Art. 22 GDPR – Automated individual decision-making, including](#)



California's CCPA was not modeled after the GDPR

Art. 22 GDPR

Automated individual decision-making, including profiling ²⁰¹⁶

1. The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her.
2. Paragraph 1 shall not apply if the decision:
 - (a) is necessary for entering into, or performance of, a contract between the data subject and a data controller;
 - (b) is authorised by Union or Member State law to which the controller is subject and which also lays down suitable measures to safeguard the data subject's rights and freedoms and legitimate interests; or
 - (c) is based on the data subject's explicit consent.
3. In the cases referred to in points (a) and (c) of paragraph 2, the data controller shall implement suitable measures to safeguard the data subject's rights and freedoms and legitimate interests, at least the right to obtain human intervention on the part of the controller, to express his or her point of view and to contest the decision.
4. Decisions referred to in paragraph 2 shall not be based on special categories of personal data referred to in [Article 9\(1\)](#), unless point (a) or (g) of [Article 9\(2\)](#) applies and suitable measures to safeguard the data subject's rights and freedoms and legitimate interests are in place.

In the United States

116TH CONGRESS
1ST SESSION

H. R. 2231

To direct the Federal Trade Commission to require entities that use, store, or share personal information to conduct automated decision system impact assessments and data protection impact assessments.

(2) AUTOMATED DECISION SYSTEM IMPACT ASSESSMENT.—The term “automated decision system impact assessment” means a study evaluating an automated decision system and the automated decision system’s development process, including the design and training data of the automated decision system, for impacts on accuracy, fairness, bias, discrimination, privacy, and security that includes, at a minimum—

(A) a detailed description of the automated decision system, its design, its training, data, and its purpose;

(B) an assessment of the relative benefits and costs of the automated decision system in light of its purpose, taking into account relevant factors, including—

(i) data minimization practices;

(ii) the duration for which personal information and the results of the automated decision system are stored;

(iii) what information about the automated decision system is available to consumers;

(iv) the extent to which consumers have access to the results of the automated decision system and may correct or object to its results; and

(v) the recipients of the results of the automated decision system;

(C) an assessment of the risks posed by the automated decision system to the privacy or security of personal information of consumers and the risks that the automated decision system may result in or contribute to inaccurate, unfair, biased, or discriminatory decisions impacting consumers; and

(D) the measures the covered entity will employ to minimize the risks described in subparagraph (C), including technological and physical safeguards.

Did not receive a vote in Congress.
Bill Introduced was 2019

2019

European Union Study



A governance framework for algorithmic accountability and transparency

This study develops policy options for the governance of algorithmic transparency and accountability, based on an analysis of the social, technical and regulatory challenges posed by algorithmic systems. Based on a review and analysis of existing proposals for governance of algorithmic systems, a set of four policy options are proposed, each of which addresses a different aspect of algorithmic transparency and accountability: 1. awareness raising: education, watchdogs and whistleblowers; 2. accountability in public-sector use of algorithmic decision-making; 3. regulatory oversight and legal liability; and 4. global coordination for algorithmic governance.

Background

Google has long championed AI. Our research teams are at the forefront of AI development, and we've seen firsthand how AI can enable massive increases in performance and functionality. AI has the potential to deliver great benefits for economies and society — from improving energy efficiency and more accurately detecting disease, to increasing the productivity of businesses of all sizes. Harnessed appropriately, AI can also support fairer, safer and more inclusive and informed decision-making. We are keen to ensure that everyone and every business can benefit from the opportunities that AI creates.

AI will have a significant impact on society for many years to come. That's why we established our AI Principles (including applications we will not pursue)¹ to guide Google teams on the responsible development and use of AI. These are backed by the operational processes and structures necessary to ensure they are not just words but concrete standards that actively impact our research, products and business decisions to ensure trustworthy and effective AI application.

But while self-regulation is vital, it is not enough. Balanced, fact-based guidance from governments, academia and civil society is also needed to establish boundaries, including in the form of regulation. As our CEO Sundar Pichai has noted, AI is too important not to regulate. The challenge is to do so in a way that is proportionately tailored to mitigate risks

The EU Artificial Intelligence Act

2025-

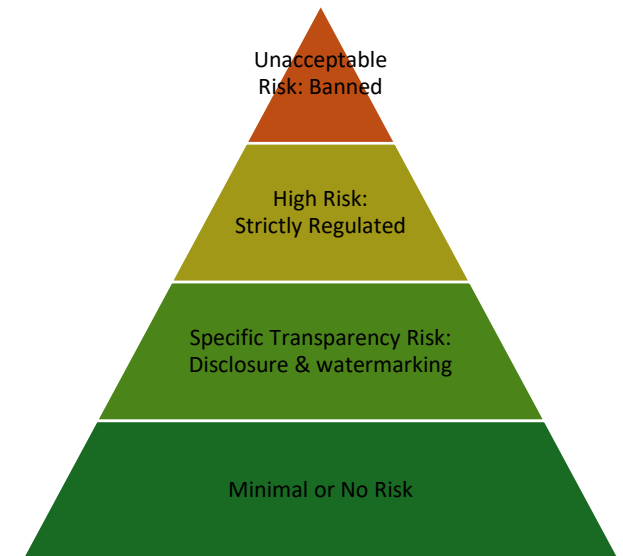
World's first comprehensive legal framework for artificial intelligence.

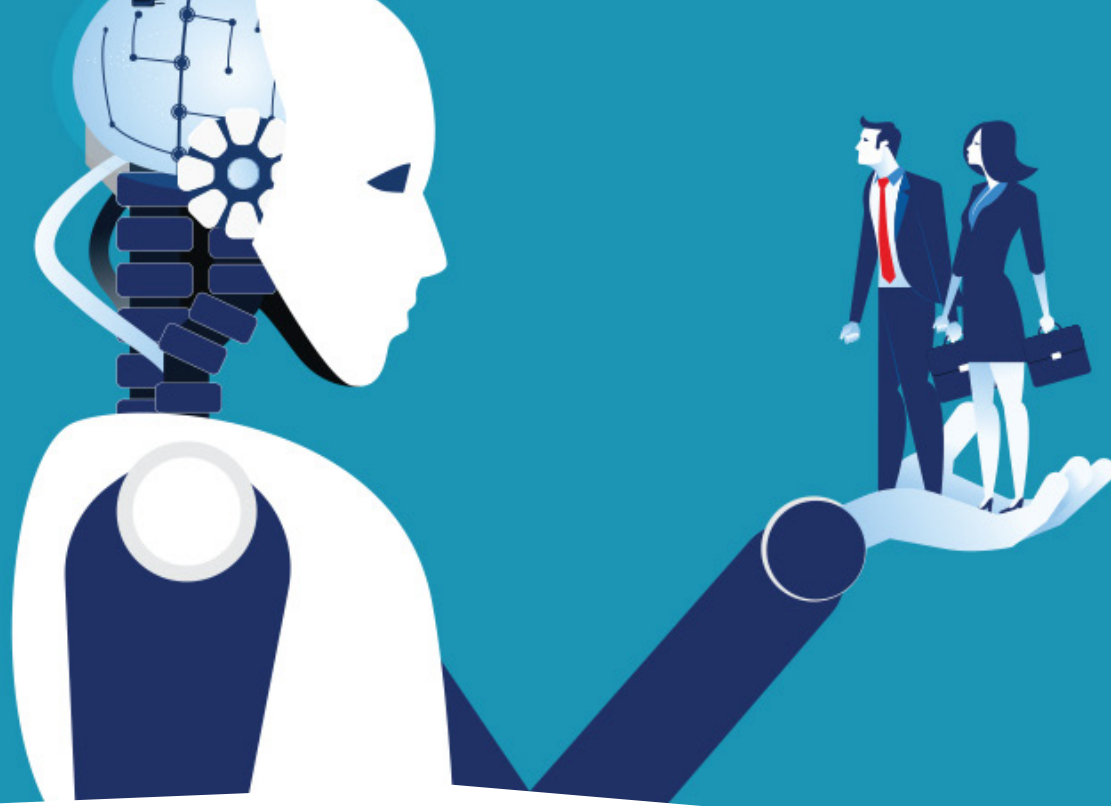


EU Artificial Intelligence Act

- **Risk-based classification of AI system**

- **Unacceptable Risk (Banned):** Systems that threaten human safety, livelihoods, or fundamental rights.
- **High Risk (Strictly Regulated):** Systems used in critical areas such as healthcare, recruitment, law enforcement, and critical infrastructure. Require: Rigorous conformity assessments, ensure human oversight, and guarantee cybersecurity.
- **Specific Transparency Risk:** AI systems that interact directly with humans (e.g., chatbots) or generate synthetic media (e.g., deepfakes, manipulated audio). Require: disclosure and watermarking
- **Minimal or No Risk:** Majority of AI systems. No binding obligations, though developers can voluntarily adopt codes of conduct.





AI Governance: Risk Mitigation

Fairness: Algorithmic Bias

“**Algorithmic bias** describes systematic and repeatable errors in a computer system that create unfair outcomes, such as privileging one arbitrary group of users over others.” [Wikipedia]



Pre-existing bias

Social and institutional norms influence design and training data choices.

Example: Evaluate job applicants for a job which is historically almost exclusively held by males.



Technical bias

Limitations of a program or computational power.

Example: Algorithm uses data that is easy to obtain but biased.



Emergent bias

Use and reliance on algorithms across new or unanticipated contexts.

Example: Use a large language model to support job candidate selection.

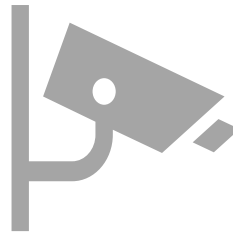
We need a mitigation plan for this risk!

Types of AI Safety

“Prevent accidents, misuse, or other harmful consequences of AI.”



AI Testing



Monitoring AI



Adversarial
robustness

How should this be ensured?

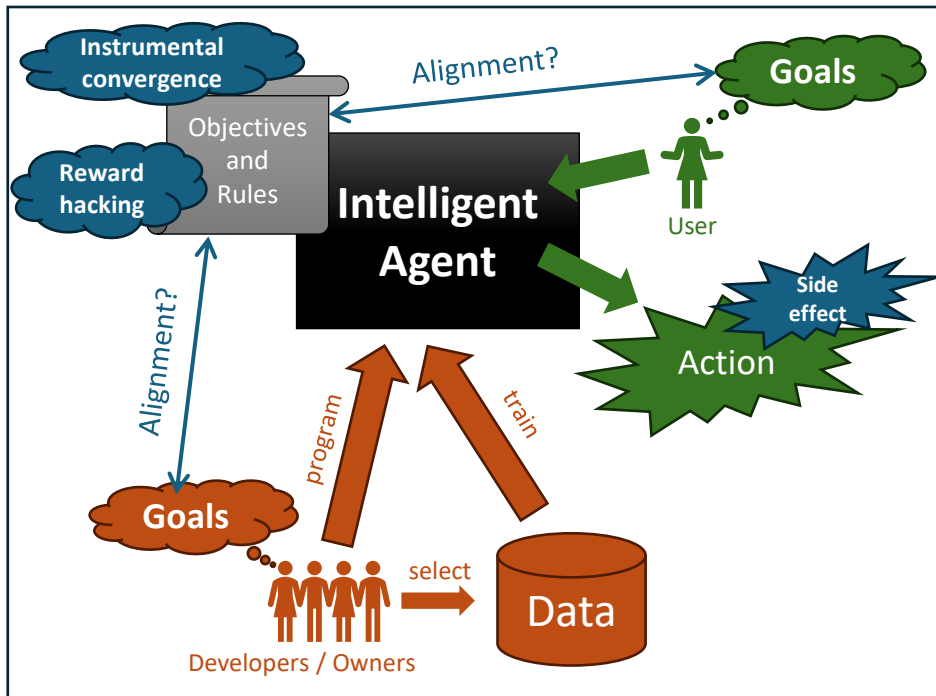
- Corporate self-regulation
- Private watchdogs
- Government action
- International treaties

Is it reasonable to assume that a superintelligent AI can get around being tested and monitored?

AI Safety vs Optimizers

Intelligent Agents are “optimizers!”

- **Goal/reward alignment:** How do we specify a robust objective function? Whose objectives are used?
- **Reward hacking:** The AI learns to exploit unintended side effects to get a high “score” without solving the objective. AI needs to follow social norms.
- **Instrumental convergence:** All intelligent agents will pursue common subgoals like the need for more power to get better at reaching its objectives. How will this need be balanced with human’s needs?



Credit: Terminator 3: Rise of the Machines. Warner Bros.

Risks of Cognitive Offloading



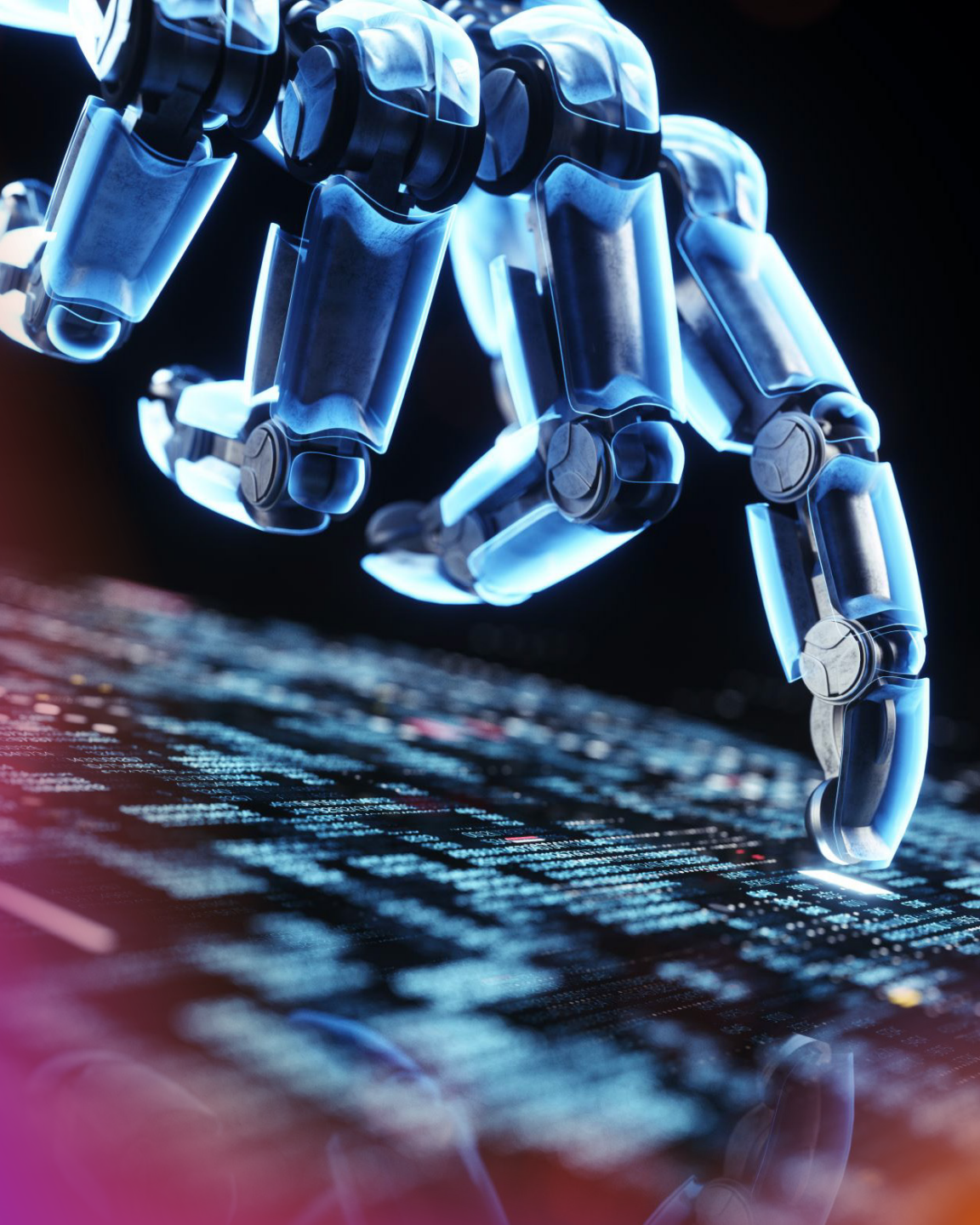
Overuse of Generative AI for **cognitive offloading** erodes fundamental critical thinking, problem-solving, and domain-specific expertise.

It poses potential risks to humans:

- **Deskilling:** Asking (current) AI for decisions leads to lower-quality decisions and erosion of skills.
- **Never-skilling:** Students use AI to by-pass learning how to research, build an argument, or develop deep subject knowledge.

Directed Co-creation*: Using AI selectively as a tool where the human structures and controls the interaction with AI can lead to **upskilling!**

* Randazzo et al, [Cyborgs, Centaurs and Self-Automators: The Three Modes of Human-GenAI Knowledge Work and Their Implications for Skilling and the Future of Expertise](#). The Wharton School Research Paper, Harvard Business School Working Paper No. 26-036



Outlook

AI is a technology that is on the verge of significant leaps...

- New technologies have always had a profound impact on the way we live and work (e.g., electricity, the internet, mobile communication).
- We can expect unprecedented gains in productivity from better **narrow AI**.
- New technologies always also present **dangers** and require governance and regulation.

This course will introduce the main techniques to create intelligent agents that act rationally.